

OSBORNE V. LOPES

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SUMMARY

Driven Data Science graduate with hands-on experience in statistical modeling, machine learning, and large-scale analytics using Python, SQL, and AWS. Experienced in building scalable pipelines and leveraging Machine Learning to generate actionable insights.

EDUCATION

Northeastern University, USA

Dec 2025

Master of Science, Information Systems

- **Coursework:** Big Data and Intelligent Analytics, Data Architecture & Business Intelligence, Data Science Methods & Tools

University of Mumbai, India

Jun 2021

Bachelor of Technology, Information Technology

- **Coursework:** Data Mining and Business Intelligence, Database Management, Cloud Computing Services

SKILLS

- **Databases & Cloud:** Snowflake, SQL, MongoDB, DynamoDB
- **Languages:** Python, R, SQL, Pro*C
- **Libraries:** BeautifulSoup, FastAPI, Pydantic, Presidio, Matplotlib, Seaborn, Numpy, Pandas, SQLAlchemy
- **Machine Learning:** Regression, Classification, Clustering, Forecasting, PCA, Tensorflow, NLP, Deep Learning
- **Analytical Toolkit:** Tableau, PowerBI, Google Data Studio, Excel, QuickSight, Athena
- **Tools:** Airflow, Streamlit, AWS, Apache Spark, Docker, Hive, EMR, Kubernetes, ADF, Control-M, RAG

EXPERIENCE

Atlas SP

Jan 2025 - Aug 2025

Data Science Co-op

New York

- Designed AWS Glue ETL pipelines to ingest & transform large Bloomberg datasets into a centralized Amazon S3 data lake, executing stored procedures to populate presentation tables in Snowflake, enabling Tableau to access up-to-date data
- Developed stress-testing models using Monte Carlo simulations and GARCH to assess tail risk and capital adequacy, scaling large-scale scenario analysis using Numpy and Scipy to evaluate 10,000+ stress scenarios and 5 TB of market data.
- Built and optimized Amazon Redshift schemas and queries for analytics, applying performance tuning techniques to reduce query latency and cost cutting average query latency by 30%
- Engineered serverless Athena workflows and QuickSight reports, delivering near-real-time insights to stakeholders.
- Automated model training, evaluation, and data refresh workflows using AWS Step Functions and event-driven triggers, reducing manual intervention by 10+ hours per week and ensuring consistent model performance.

Ingram Micro

Oct 2022 - Aug 2023

Data Scientist

Mumbai, India

- Engineered Fault-tolerant ETL pipelines to automate tasks in IMFIRST and improved workflows, boosting efficiency by 30%
- Engineered features and evaluated models using precision, recall, and ROC-AUC, improving identification of high-risk shipments
- Conducted A/B testing on reporting logic and alert thresholds using SQL queries, which increased issue detection accuracy
- Developed QuickSight dashboards, enabling operations teams to monitor shipment status, transit delays, and SLA adherence
- Improved data freshness and reporting reliability by applying hypothesis testing and statistical validation to detect anomalies
- Built feature-ready SQL datasets in Dockerized environments for regression and classification models, enabling analysis of delay patterns and prediction of SLA breach likelihood, which helped the team prioritize interventions

ICICI Securities

Jun 2021 - Sep 2022

Associate Software Developer

Mumbai, India

- Led 5 Disaster Recovery (DR) drills, by analyzing failover results ensuring uninterrupted connectivity to NSE and BSE platforms
- Maintained fresh Snowflake views driven by business logic in stored procedures, ensuring up-to-date datasets for reporting
- Utilized Python and PySpark within Azure Synapse Analytics to transform and analyze training data from multiple sources, enabling faster model training and more reliable predictions
- Deployed Python scripts for data cleaning, reducing processing time by 50% & automated Tableau reports, saving 8 hours weekly
- Addressed and resolved 20+ live issues during market hours, maintaining uninterrupted operations of critical systems

PROJECTS

Drug Review Analysis for Patient Condition Diagnosis

Sep 2023

- Crafted an NLP tool for classifying patient conditions from drug reviews, implementing data lemmatization & stemming
- Visualized sentiment distribution with pie charts to enhance data interpretation & tested various models achieving high accuracy

NewsScope

Apr 2024

- Built NewsScope, an industry-specific news aggregator, using Python (BeautifulSoup & Scrapy), Snowflake & DBT
- Designed a user-friendly Streamlit interface & automated ETL pipelines with Airflow, containerizing the platform with Docker

HR Analytics

Jul 2023

- Created an HR analytics dashboard for visualizing and analyzing employee performance and HR metrics
- Cleaned and prepared data in Microsoft Excel & designed interactive dashboards with advanced DAX calculations